

KUWAIT INTL, Kuwait

WMO: 405820

Lat: 29.227N

Lon: 47.969E

Elev: 63

StdP: 100.57

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	5.9	-10.8	1.5	20.1	-8.2	1.9	23.0	11.2	16.8	10.2	16.4	2.1	300	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	48.1	21.2	47.1	20.9	46.1	20.7	28.4	35.1	26.7	35.0	25.1	35.1	6.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.6	33.2	24.5	19.6	32.0	22.1	16.9	30.7	92.4	34.7	84.4	35.0	76.9	35.0	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	10.0	9.1	DB	1.4	50.0	1.8	0.7	0.1	50.5	-1.0	50.9	-2.0	51.3	-3.3	51.8	(4)
(5)			WB	-1.1	30.2	2.2	1.4	-2.7	31.2	-3.9	32.0	-5.2	32.8	-6.7	33.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	13.6	15.8	20.5	26.3	32.9	37.3	38.9	38.5	34.8	29.0	20.6	15.1	(6)
(7)		DBStd	9.59	2.90	3.11	3.19	3.25	2.75	2.02	1.85	1.85	2.40	2.77	3.70	3.14	(7)
(8)		HDD10.0	7	4	1	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD18.3	371	148	82	14	1	0	0	0	0	0	0	18	108	(9)
(10)		CDD10.0	6210	117	162	324	489	710	821	895	882	743	589	319	159	(10)
(11)		CDD18.3	3532	2	9	80	240	452	571	637	624	493	331	87	7	(11)
(12)		CDH23.3	58495	8	61	731	2852	7363	10475	11948	11473	8379	4396	764	45	(12)
(13)		CDH26.7	42241	1	10	241	1444	5058	8083	9468	8994	6076	2607	256	3	(13)
(14)	Wind	WSAvg	4.1	3.5	3.9	4.0	4.2	5.4	5.3	4.5	4.0	3.4	3.4	3.3	(14)	
(15)	Precipitation	PrecAvg	111	23	11	16	10	3	0	0	1	2	17	21	(15)	
(16)		PrecMax	209	95	71	67	44	29	1	1	14	38	116	67	(16)	
(17)		PrecMin	36	1	0	0	1	0	0	0	0	0	0	0	(17)	
(18)		PrecStd	46	23	16	18	11	6	0	0	3	7	28	22	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	27.9	35.7	40.0	45.2	48.9	49.5	49.1	47.0	42.0	34.2	26.6	(19)
(20)			MCWB	13.6	14.5	16.3	18.4	19.9	21.1	21.3	21.3	21.0	19.5	17.5	14.6	(20)
(21)		2%	DB	22.0	25.0	31.8	37.1	43.8	47.1	48.1	48.0	45.2	40.0	31.8	24.7	(21)
(22)			MCWB	13.7	13.4	15.7	18.1	19.8	20.5	21.2	21.3	20.6	19.2	17.8	15.0	(22)
(23)		5%	DB	20.6	23.2	29.1	35.2	42.1	46.0	47.1	47.0	44.0	38.2	29.5	22.9	(23)
(24)			MCWB	13.4	13.2	14.9	17.7	19.3	20.0	20.8	21.2	20.3	19.1	17.3	14.9	(24)
(25)		10%	DB	19.2	22.0	27.1	33.2	40.8	44.9	46.1	46.0	42.6	36.9	27.8	21.1	(25)
(26)			MCWB	13.0	13.0	14.3	17.0	19.1	19.7	20.6	21.1	19.9	18.8	17.2	14.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.3	18.4	20.2	22.1	24.7	26.1	28.7	30.7	29.2	26.9	23.9	19.9	(27)
(28)			MCDWB	20.2	21.5	24.9	29.5	32.5	36.1	36.2	35.8	33.8	30.6	27.3	22.0	(28)
(29)		2%	WB	16.8	17.4	19.0	21.0	22.9	23.2	26.6	29.1	27.4	25.8	22.5	18.7	(29)
(30)			MCDWB	19.3	20.6	23.8	28.9	33.2	39.2	37.8	35.6	33.4	30.4	26.3	21.0	(30)
(31)		5%	WB	15.5	16.1	18.0	20.1	21.6	21.7	24.4	27.3	25.4	24.4	20.9	17.3	(31)
(32)			MCDWB	18.3	19.7	23.3	28.7	34.5	42.2	39.4	36.1	34.3	29.8	25.4	20.2	(32)
(33)		10%	WB	14.2	15.0	16.9	19.3	20.7	20.8	22.6	25.0	23.0	23.0	19.4	15.8	(33)
(34)			MCDWB	17.7	18.9	23.3	28.5	35.7	42.1	42.3	38.4	36.5	29.4	24.4	19.4	(34)
(35)	Mean Daily Temperature Range	MDBR		10.5	11.5	12.6	12.8	13.9	14.9	14.8	15.4	15.7	14.6	11.8	10.8	(35)
(36)		5% DB	MCDBR	11.8	13.2	15.7	15.2	16.0	16.3	15.7	16.3	16.8	16.8	14.4	13.3	(36)
(37)			MCWBR	6.2	6.1	6.2	5.4	5.4	5.1	5.1	5.6	6.0	6.9	7.1	6.7	(37)
(38)		5% WB	MCDWB	8.9	9.3	10.6	11.4	13.0	15.7	15.0	14.5	14.6	13.3	10.7	8.6	(38)
(39)			MCWBR	5.8	5.6	5.7	5.0	5.6	6.0	7.6	8.1	7.9	7.2	6.8	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.506	0.553	0.608	0.695	0.697	0.653	0.672	0.623	0.569	0.570	0.544	0.507	(40)
(41)		taud		1.827	1.733	1.618	1.515	1.517	1.563	1.558	1.648	1.741	1.741	1.784	1.836	(41)
(42)		Ebn at Noon		696	704	699	658	660	687	672	701	720	679	654	669	(42)
(43)		Edh at Noon		178	213	254	291	291	277	277	251	221	208	185	170	(43)
(44)	All-Sky Solar Radiation	RadAvg		3.45	4.29	5.41	5.90	6.79	7.57	7.26	6.91	6.15	4.89	3.59	3.17	(44)
(45)		RadStd		0.24	0.28	0.24	0.31	0.30	0.26	0.19	0.15	0.20	0.20	0.28	0.28	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+107
(47) Regional (11 neighbors)		+0.33	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	+94	+107

Nomenclature: See separate page